High-resolution velocity modeling by seismic-airborne TEM joint inversion: A new perspective for near-surface characterization

Daniele Colombo[1], Gary McNeice[1], Diego Rovetta[1], Ernesto Sandoval-Curiel[1], Ersan Turkoglu[1], and Armando Sena[1]

Abstract

We discuss the use of helicopter-borne transient electromagnetic (HTEM) for a high-resolution spatial and vertical characterization of the near surface in a structure-controlled wadi in central Saudi Arabia. In this area, seismic data quality is poor, and seismic imaging suffers from a combination of scattering effects — due to swarms of faults reaching the surface — and large velocity variations occurring along subvertical boundaries between the wadi sediment infill and the surrounding carbonate plateaus. HTEM was selected from a suite of nonseismic methods for multiparameter velocity-model building to enhance the velocity estimation for the wadi and surrounding areas. HTEM data were modeled by performing spatially constrained 1D resistivity inversion to obtain a high-resolution image of the near surface with sensitivity to a depth of 400–500 m from the surface. Sharp boundaries of the wadi and fine vertical layering, obtained from the HTEM inversion, provide detailed information about the parameter variations in the near surface. A seismic-HTEM joint-inversion approach is developed using a cross-gradient structural operator to constrain the velocity inversion with the higher resolution HTEM data. Joint-inversion results provide sharp velocity reconstruction across the wadi boundaries and increase the dynamic range of the velocity variations when compared to a single-domain tomographic approach in which the HTEM contribution is ignored. Superior imaging results in both time and depth are derived from velocities estimated by the seismic-TEM joint-inversion approach.

Introduction

Oil and gas exploration in the Middle East has shifted from major structures (e.g., most supergiant fields) to stratigraphic traps and low-relief structures. These prospects can be a few kilometers long and present closures with an expression of a few tens of milliseconds (e.g., 20–30 ms) in the time domain or 30–40 m relief in the depth domain. Such features can be generated easily on a seismic time section by poorly corrected near-surface anomalies such as those introduced by karsts or collapse structures, dunes, and wadis (Figure 1). Accurate near-surface velocity modeling and corrections become the single most important step for correct seismic imaging to reduce exploration risk. Emphasis is placed on long-wavelength near-surface features that often cannot be resolved with conventional 3D seismic acquisition geometries that severely undersample the near surface. In addition to the low fold at shallow depths, production seismic also suffers from poor sampling in dune and farmed areas where vibrators typically cannot operate. Presence of scatterers and velocity inversions in the near surface add to the problem and often undermine the capability of conventional seismic data to effectively image shallow anomalies. Selected high-resolution nonseismic methods can complement seismic effectively in the characterization of the near surface by filling the gaps and enhancing sensitivity to geologic features that are not properly sampled by seismic methods (e.g., low-velocity features and velocity inversions). Examples are the high horizontal resolution of precision gravity methods to map voids or low-density materials (e.g., dunes), and the sensitivity of time/frequency electromagnetic methods to map resistivity variations, where velocity inversions are typically transitions from high- to low-resistivity formations (high velocity to low velocity).

The practical implementation of high-resolution nonseismic methods in the context of seismic exploration requires the following conditions: dense spatial sampling (comparable or superior to seismic), efficient and robust quantitative integration with seismic, speed of execution, and cost management (e.g., nonseismic methods have a complementary nature and must represent a fraction of the seismic total cost). Such requirements are typically met by airborne systems and, for the specific problem of near-surface investigation, by helicopter-borne systems. By combining low flying altitude and low cruising speed, these methods are able to provide the required spatial sampling, sensitivity to small anomalies, and accuracy of measurements needed for accurate near-surface characterization.

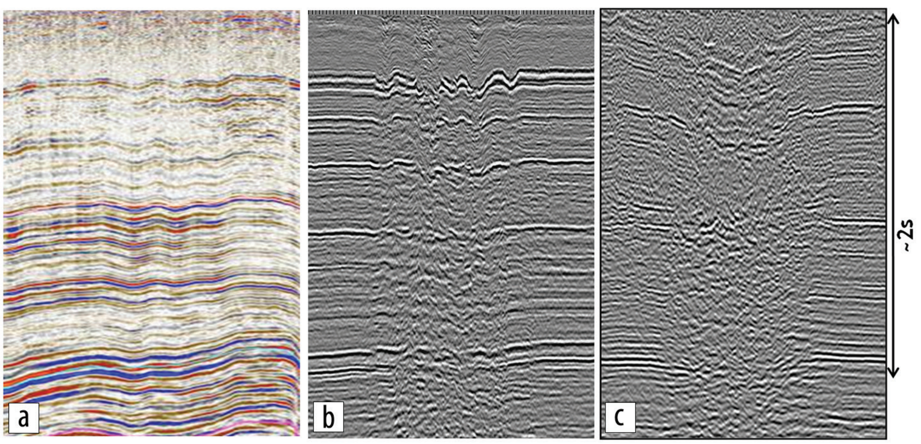

Figure 1. Examples of long wavelength distortions in land seismic data in Saudi Arabia after statics corrections, including (a) sand dune distortions, (b) karsting at around 500 m in depth, and (c) the effect of a deep wadi (example discussed in the present paper).

Airborne electromagnetic (AEM) systems are widely applied in mineral and groundwater exploration and are capable of acquiring several thousand kilometers of flying lines in a short time with exceptional spatial sampling. In the early 2000s, helicopter transient EM systems (HTEM) were introduced and have developed rapidly over the past 15 years (see Allard, 2007, for an overview of the available systems and their development history). Five years after the introduction of HTEM,

[1]Geophysics Technology, EXPEC Advanced Research Center, Saudi Aramco.

http://dx.doi.org/10.1190/tle35110977.1.

advancements such as increased source moment and reduced system noise, along with their superior terrain contouring and lower cost, made these the most commonly flown AEM systems (Nabighian and Macnae, 2005). The introduction of dual-moment sources (Sørensen and Auken, 2004; Chen et al., 2013) has given HTEM systems the increased bandwidth to replace ground TEM investigations in many situations. The increased data density, uniform spatial sampling, and data consistency have reduced the need for ground follow-up (Taylor, 2005). Helicopter-borne gravity gradiometry systems are also available (Boggs et al., 2007). Various efforts have been made to develop a new generation of gradiometers able to reach a 1 Eö repeatability accuracy (Brian Main, personal communication, 2016), which will approach the accuracy provided by ground-based gravity measurements. The availability of reliable and accurate EM and gravity helicopter-borne methods opens the road to a new set of applications important for near-surface characterization.

Having addressed the requirements of cost management and of dense and accurate measurements, the next challenge is provided by the establishment of robust processing and integration of multiphysics data. Quantitative, semiautomatic joint-inversion techniques (Colombo and De Stefano, 2007; Colombo et al., 2013) are needed, especially for 3D data sets, to enable a unified approach of seismic and nonseismic methods for near-surface analysis. Preprocessing of the nonseismic data should be fast and reliable. A fraction of the time needed by the seismic data processing should be used for the preprocessing and inversion of the nonseismic data, and, more importantly, the nonseismic results should be provided in a timely manner to be used as an additional tool for the near-surface velocity modeling at the beginning of the seismic processing sequence. Joint-inversion algorithms must be robust and semiautomatic to allow execution even by geophysicists with limited experience with nonseismic methods.

To follow, we describe the acquisition of helicopter-borne TEM over a geologically complex wadi area in central Saudi Arabia characterized by high noise in the seismic data where joint inversion of TEM and seismic first-break (FB) traveltimes was performed to aid the velocity modeling of the near surface. The example represents the realization of the aforementioned conditions to provide effective multiphysics near-surface solutions for seismic imaging.

Survey area and background work

The area under investigation is located along a major structure-controlled elongated depression (wadi) known as wadi Sahba. The system is part of the southern extension of the Central Arabian Graben System delimiting the East Arabian Block to the west (Figure 2). The seismic data quality in the wadi is extremely poor due to strong scattering effects probably caused by patterns of flower faults reaching the surface (Weijermars, 1998). Other complications to seismic imaging derive from the strong velocity contrasts between soft sediments present in the wadi and carbonate formations outcropping on the surrounding plateau. The presence of farmland, where vibrators cannot operate, also produces uneven sampling and acquisition gaps inside the wadi. The wadi area itself is of great relevance from an exploration point of view as it lies at the intersection of three major reservoirs in the Mesozoic and Paleozoic sections, the extents of which are obscured by the overlying wadi structure.

Following encouraging results of 2D multiphysics and joint-inversion studies conducted in the area (Colombo et al., 2012), a comprehensive nonseismic 3D acquisition program was completed in December 2014 consisting of precision gravity on a 200 × 200 m grid, helicopter-borne TEM, and audiomagnetotellurics (AMT) soundings, acquired to extend the depth resolution of the helicopter TEM data in the wadi section. The individual multiphysics data

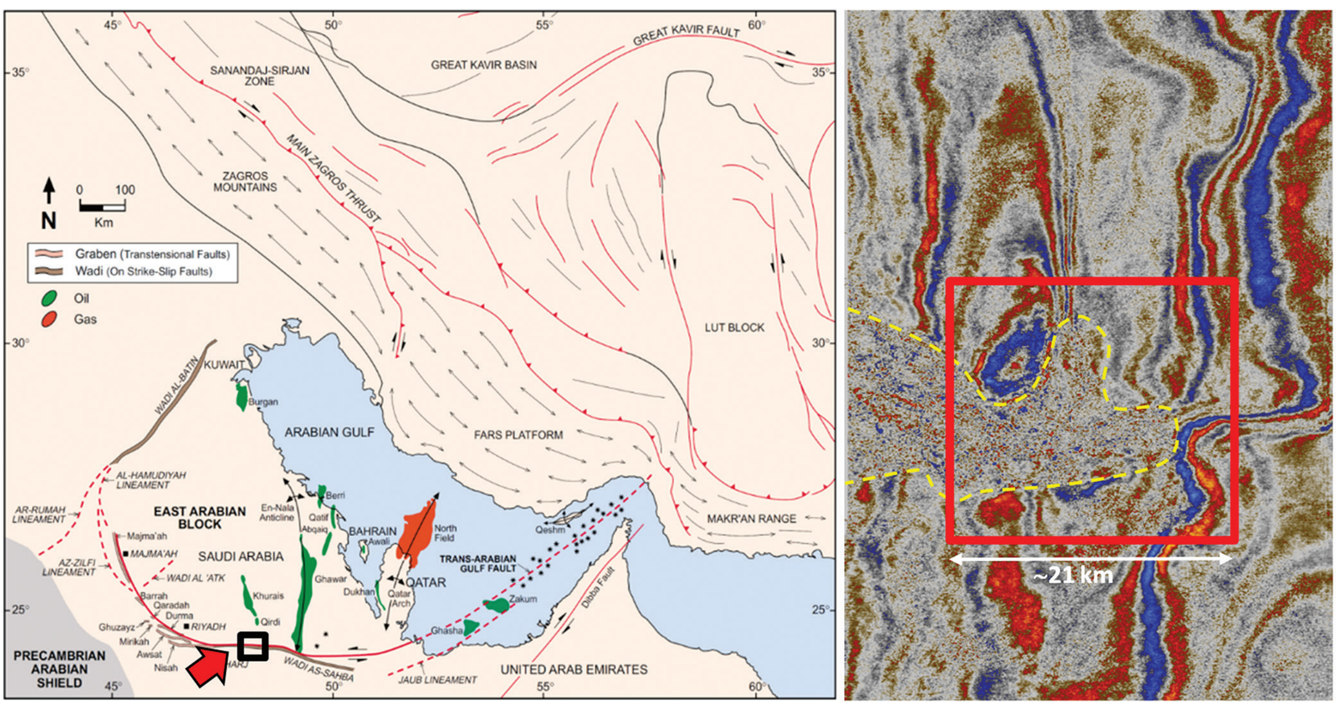

Figure 2. Structural map from Weijermars (1998) showing the location of the study area (left) and a snapshot of existing seismic (time slice at 1 s) with the outline of the wadi (dashed line) and of the 3D study detailed in the present paper (red solid line).

sets were first processed and inverted separately to optimize corrections and noise removal, and to analyze the robustness of the different data sets and the consistency of the structural information they could provide. The results of such analysis provided evidence of strong structural similarity between the inverted resistivity and density and the existing seismic reflectivity and seismic tomography results (Colombo et al., 2015). The consistency of the information across different parameter domains and the resolution of the nonseismic data provided the necessary background for a quantitative integration approach based on multiparameter joint inversion. The structural similarity with seismic and the dense spatial sampling of the helicopter-borne TEM data suggested they were the best candidates for joint inversion whereby we aim to capture the high-resolution structure details from resistivity measurements into the poorly resolved and noise-affected P-wave velocity model from inversion of FB traveltimes.

Helicopter-borne TEM

Among the different nonseismic data sets acquired in the study area, the HTEM (Chen et al., 2014) is the most promising technique for high-resolution characterization of the near surface at an effective cost/benefit ratio. HTEM data were acquired over the survey area with north-south fly lines spaced 100 m apart. The acquisition system was designed to provide adequate depth of penetration by ensuring large transmitted current and dipole moment with suitable recording time (Table 1). The low flight altitude of nominally 30 m and the relatively low flying speed ensured collection of a rich data set corresponding to an output-stacked TEM sounding acquired every 2.7 m along the flight lines for a total of about 1.6 million TEM soundings for the surveyed area. The availability of ground-based measurements from a former 2D multiphysics pilot test (Colombo et al., 2012) enabled comparison of the ground-based with the corresponding helicopter-borne TEM soundings.

Table 1. Helicopter-borne TEM (HTEM) acquisition parameters.

TEM acquisition parameters	
Recording time windows	15 ms
Transmitter current	1200 A
Dipole moment	1.7x10^6 A.m^2
Stacked TEM curve spacing	~ 2.7 m (inline)
Transmitter waveform	4 ms, 30 Hz, half-sine
Fly-line spacing	100 m
Total soundings	~ 1.6 Million

The airborne system was flown along one line where ground-based TEM data were available. Soundings at corresponding locations were compared after performing an apparent resistivity transform (Figure 3). Results indicate overall consistency of the ground data and airborne data, with the former displaying usable measurements at shorter time intervals and the latter extending the data to the late times, often by one decade. HTEM data typically display more stable results than the corresponding ground-based TEM soundings, probably because of a more stable geometry of the transmitter-receiver loops. Current HTEM systems can be equipped with dual-moment transmitters allowing coverage of the early times together with the deep time responses to further enhance resolution of very shallow features. The combination of frequency-domain and time-domain helicopter-borne acquisition allows further expansion of the sensitivity to provide even broader bandwidth of electromagnetic responses for virtually no limitations in the resolution by which the near surface can be characterized.

Previous work on multiphysics velocity modeling and prestack depth migration on pilot 2D lines (Colombo et al., 2012) suggested the presence of deep roots below the wadi where the velocity anomalies associated with the wadi extend for several hundred meters below the surface. These considerations guided the definition of parameters for the HTEM system to enable extended depth penetration without compromising the near-surface resolution (Table 1).

A concise representation of the acquired HTEM data is provided by Figure 4a in which an apparent resistivity transform from the first time window after pulse shutoff is displayed in comparison to a shallow time slice at 164 ms of the stacked seismic cube after final processing (Figure 4b). Distribution of the apparent resistivity matches structures displayed by the seismic amplitudes but at even higher resolution and without acquisition artifacts. Several details that are blurred or hidden on the seismic image become apparent on the resistivity map. The wadi area boundaries are sharply defined in the resistivity map as well as the parameter variations within the dune field located

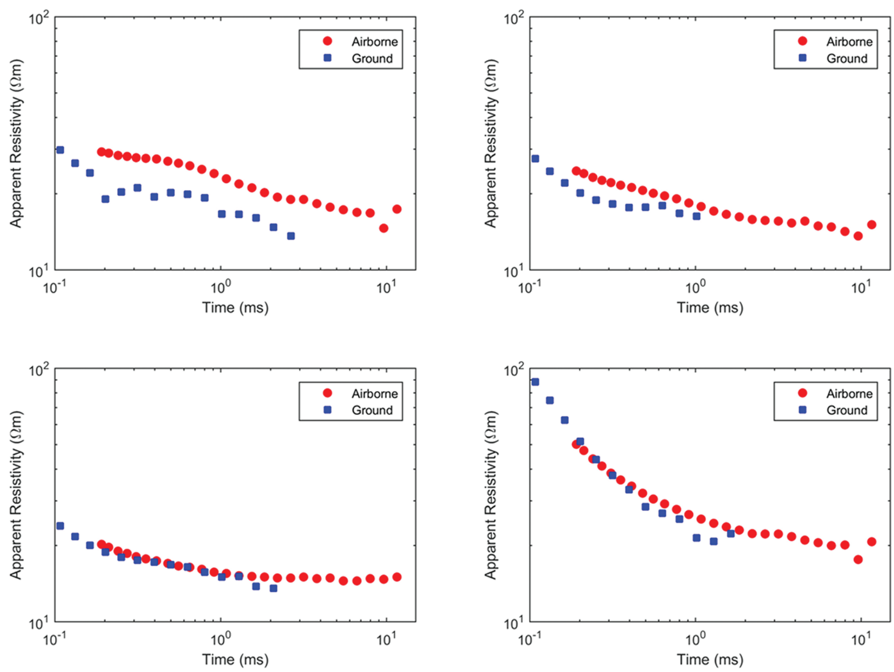

Figure 3. Comparison of helicopter-borne and ground-based TEM soundings at corresponding locations.

in the block's southeast quadrant. The area inside the wadi shows large variations of the apparent resistivity, suggesting the wadi interior structure to be more complex than a simple alluvial sediment infill. In addition, the HTEM data fill in the gaps created by surface access restrictions (e.g., farms) and dunes to complement the description of the near surface, which is largely undersampled by conventional seismic acquisition layouts.

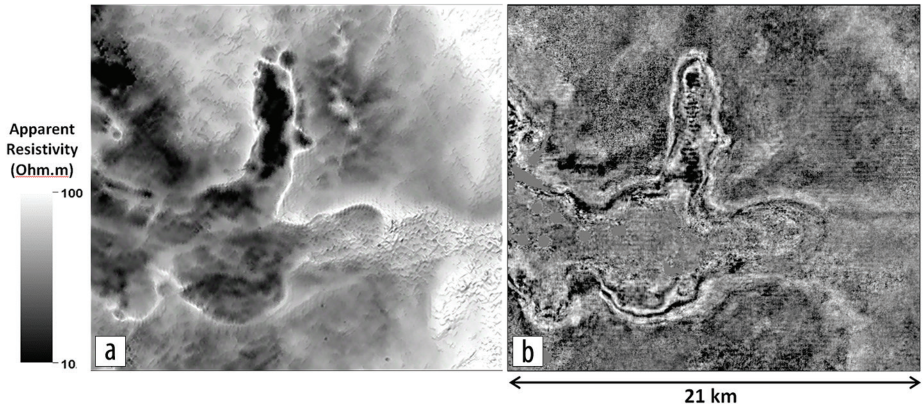

Figure 4. Comparison between (a) an apparent resistivity transform of early-time HTEM data with (b) a seismic time slice at 164 ms.

HTEM data processing consisted primarily of automatic elimination of outliers from the time-decay curves, masking of measurements above conductive infrastructure such as pipelines, tilt correction for transmitter and receiver loops, and system attitude accounted for in the forward modeling. Inversion was performed using a 1D inversion method with 3D spatial regularization (Rovetta et al., 2013). Acquisition geometry and the background horizontally layered geology suggest that a spatially constrained 1D inversion assumption is a reasonable choice (Viezzoli et al., 2008), further justified by the gain in computing speed and by the volume of data involved. Analysis of the horizontal components of the transient also indicates that the 3D effects are negligible. HTEM inversion results are shown in Figure 5, where the deep roots of the wadi become evident in an otherwise subhorizontal stratified geology.

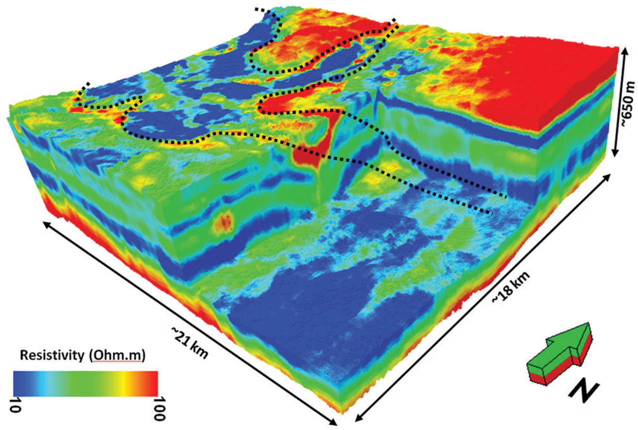

Figure 5. HTEM inversion results of the study area with the outline of the wadi indicated by a dashed line. The resistivity distribution indicates the presence of deep roots below the wadi in an otherwise horizontally layered geology.

Joint-inversion method

We focus on the derivation of a generalized joint-inversion scheme (Colombo and De Stefano, 2007) for the 3D case where each inverse problem is formulated as a constrained least-squares problem and is solved using Lagrange multipliers and a preconditioned conjugate gradient iterative algorithm to minimize the total objective functional of the form:

$$\phi_t(\boldsymbol{m}) = \phi_m(\boldsymbol{m}) + \frac{1}{\lambda_1}\phi_d(\boldsymbol{m}) + \frac{1}{\lambda_2}\phi_x(\boldsymbol{m}) + \frac{1}{\lambda_3}\phi_{rp}(\boldsymbol{m}), \quad (1)$$

where $\phi_d(\boldsymbol{m})$ is a measure of the data misfit, $\phi_m(\boldsymbol{m})$ is a model regularization term, $\phi_x(\boldsymbol{m})$ is the structural term represented by the cross gradients, $\phi_{rp}(\boldsymbol{m})$ is a rock-physics term that can be used in combination with the cross-gradient structural terms to enforce similarity among parameter distributions, and λ_i are Lagrange multipliers. The model regularization functional, $\phi_m(\boldsymbol{m})$, is defined as:

$$\phi_m(\boldsymbol{m}) = (\boldsymbol{m} - \boldsymbol{m_{ref}})^T \boldsymbol{W_m^T W_m} (\boldsymbol{m} - \boldsymbol{m_{ref}}) = \left\|\boldsymbol{W_m}(\boldsymbol{m} - \boldsymbol{m_{ref}})\right\|_{L_2}^2, \quad (2)$$

where $\boldsymbol{m}$ and $\boldsymbol{m}_{ref}$ are, respectively, the unknown and the reference models parameters (M_S slownesses and M_R resistivities); $\boldsymbol{W}_m$ is a model weighting (or covariance) matrix including prior information on the model, smoothing operators, and spatially dependent weighting functions (e.g., a 3D spatially consistent operator is used for 1D HTEM inversion). The data misfit functional $\phi_d(\boldsymbol{m})$ is defined as:

$$\phi_d(\boldsymbol{m}) = (\boldsymbol{Jm} - \boldsymbol{d_{obs}})^T \boldsymbol{W_d^T W_d} (\boldsymbol{Jm} - \boldsymbol{d_{obs}}) = \left\|\boldsymbol{W_d}(\boldsymbol{Jm} - \boldsymbol{d_{obs}})\right\|_{L_2}^2, \quad (3)$$

where $\boldsymbol{d}_{obs}$ is the vector of the observed data (N_t traveltimes and N_{em} HTEM observations); $\boldsymbol{J}$ is the Jacobian or the sensitivity matrix; and $\boldsymbol{W}_d$ is a data weighting (or covariance) matrix taking into account the relative importance of the observations and the effect of the noise in the data. An additional nonlinear transformation is applied to the model to constrain the new solution to be confined into defined physical bounds (Habashy and Abubakar, 2004). The two remaining terms, $\phi_x(\boldsymbol{m})$ and $\phi_{rp}(\boldsymbol{m})$, represent cross-gradient (Gallardo and Meju, 2004) and rock-physics misfits.

$$\phi_x(\boldsymbol{m}) = \Sigma_{k=1}^{K}\left\|\boldsymbol{x}_k\right\|_{L_2}^2. \quad (4)$$

$$\phi_{rp}(\boldsymbol{m}) = \Sigma_{k=1}^{K}\left\|R_k - f_{rp}(s_k)\right\|_{L_2}^2. \quad (5)$$

Perfect structural similarity between slowness and resistivity models is achieved when ∇s_k and ∇R_k share the same direction and $\boldsymbol{x}_k = 0$. We encourage structural similarity between the models by minimizing the cross product of the model gradients. Empirical or physical relations mapping different geophysical domains can be derived through rock physics (Carcione et al., 2007). With the exception of the rock-physics term, all other terms of the generalized joint-inversion objective function were used in the joint-inversion work for wadi Sahba.

Joint-inversion results

Among the multiphysics methods acquired in the study area, the resistivity model from the inversion of HTEM data provides the highest resolution images of the near surface in both horizontal and vertical dimensions. The precision gravity data set and the corresponding 3D density inversion (Colombo et al., 2015) also provide satisfactory spatial sampling but intrinsically suffer from poor vertical resolution. The separation of residual gravity anomalies associated with the response from the near surface from a background gravity trend is also performed through an empirical rather than a deterministic approach, leaving some degree of uncertainty. The above considerations led to the decision to focus on the joint inversion of HTEM and seismic traveltimes to try to boost the resolution of the velocity model for near-surface corrections.

Two inversion approaches were followed consisting of a cooperative joint inversion, in which the stand-alone derived resistivity distribution was used as a "reference model" through the cross-gradient minimization but was not inverted for (cooperative joint inversion), and a simultaneous joint inversion scheme, in which both data sets were simultaneously inverted subject to the cross-gradient minimization constraint. Both approaches provided satisfactory results with consistent and steady minimization of the total (1) and of the individual objective functionals (3, 4) as shown by Figure 6. Traveltime residuals were slightly smaller for the simultaneous joint inversion (root mean squared error, rms = 32.6 ms) where the HTEM resistivity was allowed to vary, when compared to the cooperative joint inversion (rms =32.8 ms) where the resistivity model was fixed. The cross-gradient misfit functional decreased in both cases with the best performance obtained by the cooperative joint-inversion case. Given that the HTEM resistivity inversion model provides the highest resolution, a decision was made to continue and finalize the joint-inversion work by means of a cooperative method in which resistivity is used as regularization to the seismic traveltime inversion through the cross-gradient functional. The described approach is easier to implement than simultaneous joint inversion and provides a flexible and automated scheme for regularization; the distribution of a different physical parameter structurally related to the distribution of the main parameter to be inverted (HTEM resistivity in our case).

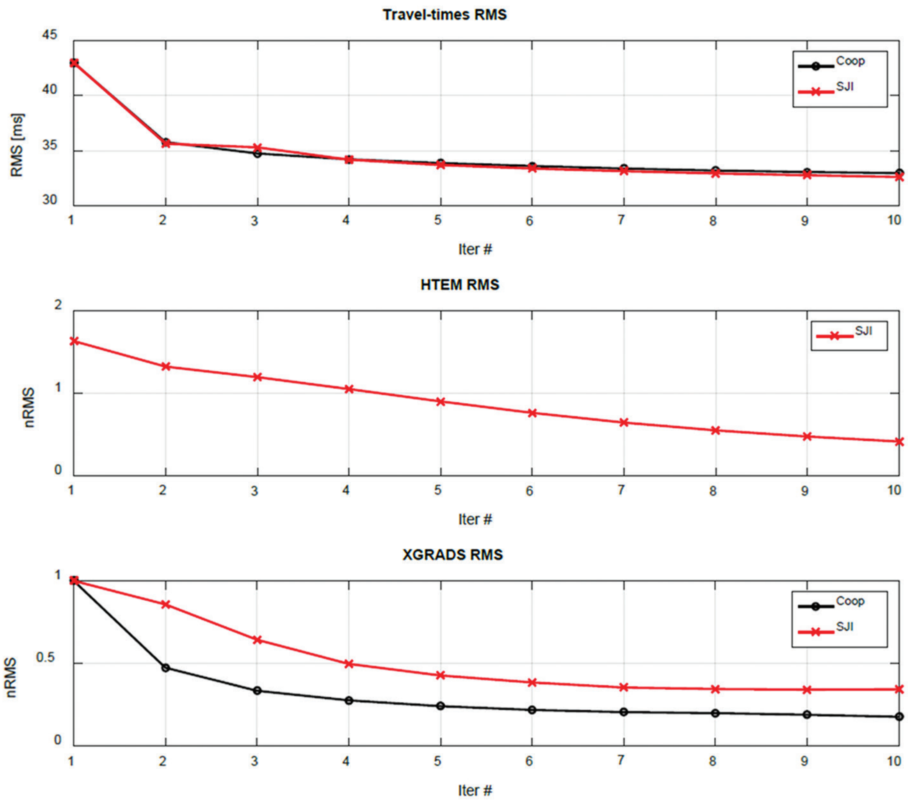

Figure 6. Comparison of cooperative and simultaneous joint-inversion (SJI) performance for seismic traveltimes and HTEM data.

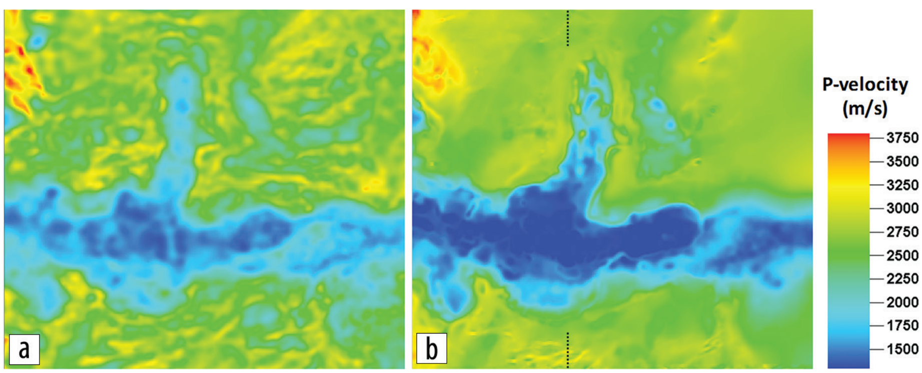

Figure 7. P-wave velocity distributions at 340 m elevation (MSL) derived from (a) stand-alone tomography and (b) joint inversion with resistivity as reference model through cross-gradient regularization. Both inversion results are shown at a common rms seismic traveltime misfit of 32.8 ms.

A shallow depth slice of the velocity field from the cooperative inversion results (Figure 7b) is compared to the velocity distribution obtained by traveltime inversion alone (Figure 7a) where the same traveltime input data set is inverted without the use of structural regularization to the resistivity model. The velocity model from joint inversion displays sharper velocity variations between the wadi and the surrounding plateau with a larger range of velocities. The velocity model from joint inversion also displays smoother, more stable velocity values in the plateau area where horizontal layering is expected. It should be noted that the minimization of the cross-gradient functional (4) corresponds to the minimization of the difference in direction of the model gradients independent of the corresponding sign. This property allows use of correlated and/or anticorrelated features in the cross-gradient calculation as long as they share common structure boundaries.

A cross section of the velocity and resistivity models (Figure 8) shows the footprint of the resistivity distribution in the joint-inversion velocity model providing overall sharper spatial resolution and larger dynamic range when compared to the results of the stand-alone traveltime inversion. This can be explained by the mechanism of regularization imposed on the noisy traveltime data that, in the case of stand-alone inversion, is provided by a Laplacian smoothness regularization acting on the velocity model and, in the case of the cooperative inversion, is provided by the cross-gradient structural constraint linking the velocity model to a high-resolution and stable resistivity distribution. The cooperative joint-inversion scheme allows us to combine at the same time an effective regularization of the noise with a boost in the model resolution.

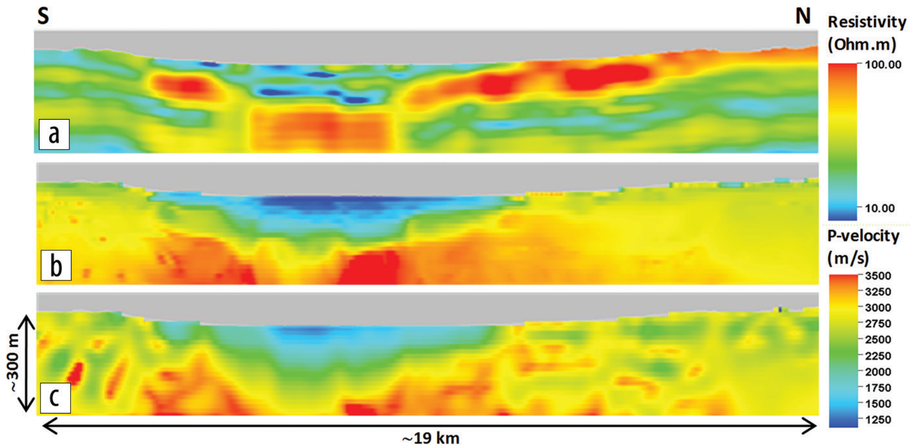

Figure 8. Cross section through the wadi area comparing (a) the resistivity model from HTEM inversion, (b) the P-wave velocity model from cooperative joint inversion of first-arrival traveltimes and HTEM resisitivity model, and (c) the P-wave velocity from stand-alone traveltime inversion.

Seismic time-domain processing

We focus on the eastern side of the survey where the boundaries of two major reservoirs are obscured by the overlying wadi structure. We evaluate the quality of the derived velocity fields on seismic imaging by computing statics corrections for the stand-alone tomographic model (Figure 9a) and for the joint-inversion model (Figure 9b). We observe that the wadi area and the dune area in the southeast quadrant show larger dynamic range for the statics corrections for the joint-inversion model. The boundaries of the wadi also provide sharp statics variations in the joint-inversion model. Statics values are applied to the traces, and a seismic stack volume is generated using a few common velocity functions from a previous project in the area. Processing of the traces is kept to an absolute minimum (amplitude balancing, deconvolution, linear noise removal) with the intention of isolating and evaluating the effects of the different statics models on the seismic data. A time slice at 1832 ms (target level) with tomostatics applied (Figure 10a) is compared to the corresponding time slice using joint-inversion statics (Figure 10b). Better closures can be observed in the wadi area by applying the statics model from joint inversion. Comparison of the seismic inline sections through the wadi shows that the joint-inversion statics (Figure 11b) provide a better solution with more continuous reflectors when compared to the tomostatics solution (Figure 11a). Both the long-wavelength component statics correction in the wadi (left arrow) and the shorter-wavelength statics component caused by the dune field (right arrow) enhance the seismic image by providing better stacking power and alignment of reflected events.

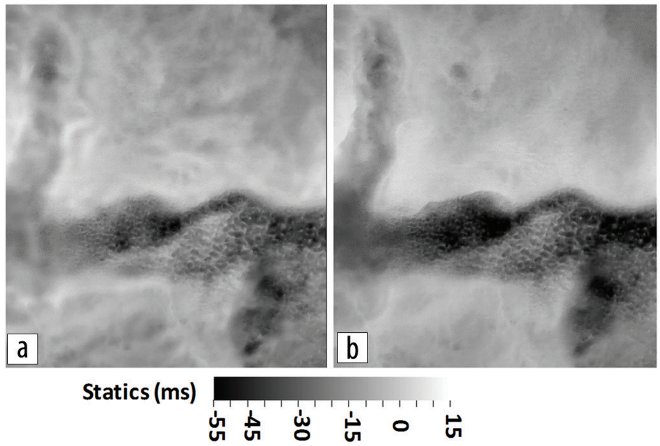

Figure 9. Statics solutions for the eastern part of the survey (a) from tomographic velocities and (b) from joint-inversion velocities.

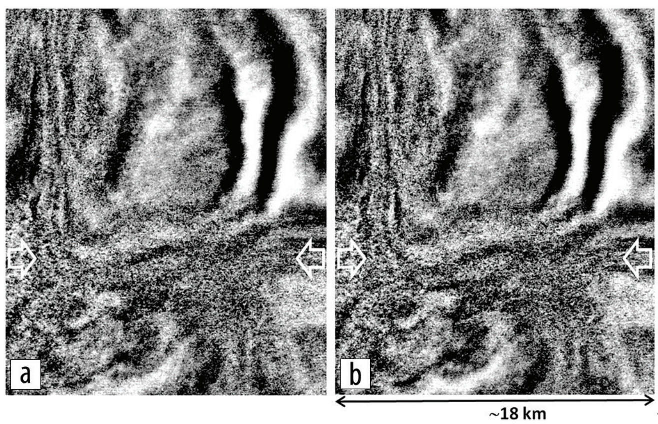

Figure 10. Time slice at 1832 ms comparing (a) tomostatics application and (b) joint-inversion statics application.

Seismic depth processing

To obtain the maximum benefit from the high-resolution near-surface velocity model, a depth-imaging workflow was applied after finalization of the time-processing step. Depth processing is highly recommended for the definition of fine structural details but can be achieved only through robust velocity modeling. For the specific case under consideration and in general for the geology characterizing most of the Arabian peninsula, the geologic complexities producing sharp velocity variations tend to be concentrated in an extensive near-surface section covering a depth range of several hundred meters, while the deep velocity structure is

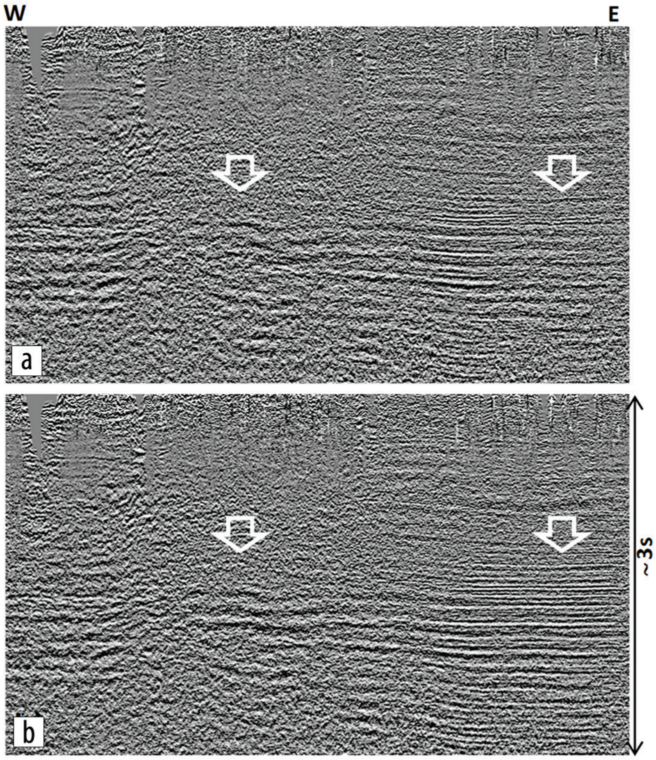

Figure 11. Inline seismic section through the wadi area with different statics models including (a) the tomostatics solution and (b) the joint-inversion statics solution.

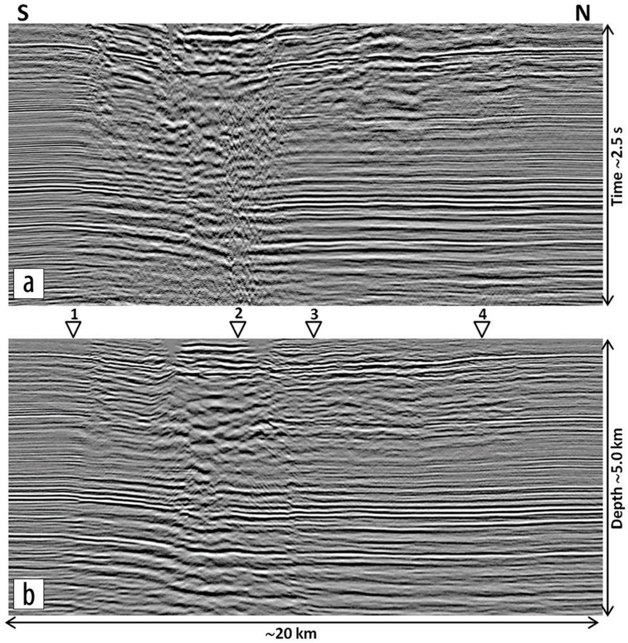

Figure 12. Crossline section through the wadi along the profile shown in Figure 8 showing (a) the reference time stack after time processing and (b) the prestack depth migration using the near-surface joint-inversion velocity model.

typically simple and easy to estimate. Depth imaging should be the preferred choice to correctly image deep reservoir structures (e.g., faults and fractures), and "deep" near-surface velocity-model-building capabilities should be provided to sustain the depth-imaging workflow.

Kirchhoff prestack depth migration was performed on the 3D seismic data, adopting the strategy of combining short/medium-wavelength statics corrections and long-wavelength velocity modeling from joint inversion to account for the "deep" near-surface section of wadi Sahba. The smooth background velocity-depth model was obtained using a Dix transform of the rms velocity-time model. This strategy allowed quick convergence to a robust velocity model for prestack depth migration. The depth-migrated seismic crossline in the center of the wadi (corresponding to the profile shown in Figure 8) is displayed in Figure 12b together with the reference time stack (Figure 12a). The robust integrated velocity modeling of the near surface associated with the depth-imaging procedure allows us to greatly improve deep reflector continuity in the wadi section and to image with high fidelity the deep structure. Inspection of common-image gathers along the crossline section (Figure 13) allows us to assess the accuracy of the derived velocity model as indicated by the flatness of the reflected events in the offset-depth domain.

Discussion

Wadi Sahba is considered one of the most challenging areas for seismic imaging in Saudi Arabia. A combination of effects, such as scattering and attenuation, severely degrade the seismic quality. Complex near-surface velocity distributions related to shallow fault structures add further challenges for seismic imaging. In this context, accurate estimation of the near-surface complexities and deep imaging challenges have been addressed by acquisition and processing of HTEM data and by development of a robust joint-inversion workflow with seismic data. The derived high-resolution velocity model was used to correct for the near surface and to enhance seismic imaging in time domain. The joint-inversion velocity model was also used for prestack depth imaging, which provided enhanced continuity of deeper structures below the wadi.

A close examination of the depth-migrated seismic in the section close to the surface is shown in Figure 14 where the HTEM resistivity model is co-rendered with the seismic cube image. Despite the lack of fold in the seismic data and the presence of noise, it is possible to appreciate the fidelity by which the HTEM resistivity model is matching the fine details provided by the depth-domain seismic volume. Faults, horst and graben-like structures, and undisturbed horizontal layering are all well imaged by the resistivity distribution from HTEM to match the structures imaged by seismic. The capability of HTEM methods to image conductors (most likely related to velocity inversions) shall be further exploited in future development of joint-inversion techniques with seismic, perhaps through exploiting seismic waveforms, to provide robust and detailed images of an extended section of the near surface to be used for seismic imaging processes in time and depth domains.

The techniques and procedures described here for the imaging of wadi Sahba contain all the elements to streamline the application of multiphysics in challenging seismic exploration areas. The requirements of dense spatial sampling, efficient and robust quantitative integration with seismic, speed of execution, and cost management are all met by helicopter-borne EM techniques and by robust joint-inversion workflows.

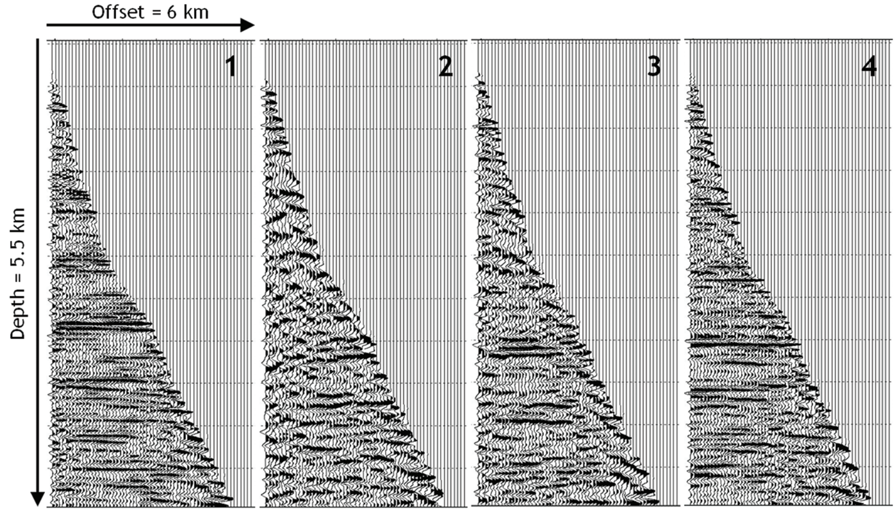

Figure 13. Common-image gathers along the crossline in Figure 12 showing the quality of the velocity field derived through seismic-HTEM joint inversion.

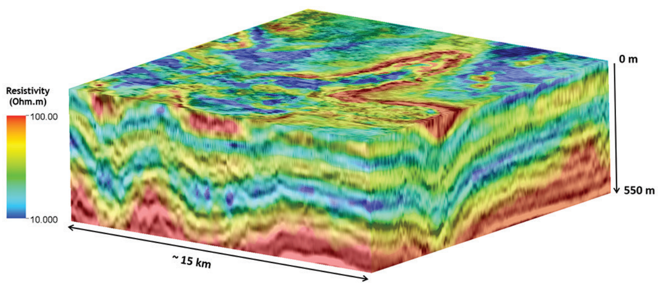

Figure 14. Seismic cube from 3D prestack depth migration with co-rendered resistivity from HTEM inversion. The resistivity distribution provides the fine description of the near-surface structures and layering as inferred from the depth-domain seismic images.

Conclusions

Near-surface velocity modeling is very challenging in arid environments typical of the Middle East. Recent exploration trends, placing emphasis on low-relief structures and stratigraphic traps, call for novel and robust solutions for the near-surface problem. Production 3D seismic utilizing sparse cross-spread acquisition geometries produce severe undersampling of the problematic near surface. Propagation of seismic waves through the near surface is also affected by additional problems related to scattering, attenuation, poor signal penetration, and velocity inversions, which produce noise and problematic analysis of first-arrival seismic phases. For these reasons, velocity-analysis workflows often are unable to resolve near-surface velocity anomalies, causing severe distortions in the seismic images. Helicopter-borne EM methods (time and frequency domain) are ideally suited to provide accurate and cost-effective near-surface characterization based on the resistivity parameter distribution. Robust joint-inversion approaches provide the tools for multiphysics data integration to obtain high-resolution velocity models of the near surface. Achievement of this goal is subject to the existence of robust and semiautomatic quantitative integration schemes such as the described joint-inversion method. The rapid evolution of airborne EM methods and of efficient quantitative multiphysics integration approaches are likely to provide the oil and gas exploration industry with a completely new range of competitive solutions to solve the near-surface problem in a variety of geologic environments.

Acknowledgments

We thank Saudi Aramco for permission to show the results detailed in the paper. Pierre Leger (Saudi Aramco Geophysical Data Processing Department) is also thanked for the work performed in the time processing of the wadi Sahba seismic data. TLE

Corresponding author: daniele.colombo@aramco.com

References

Allard, M., 2007, On the origin of the HTEM species, in B. Milkereit, *ed.*, Exploration in the new millennium: Proceedings of the Fifth Decennial International Conference on Mineral Exploration, 355–374.

Boggs, D., R. Maddever, J. Lee, R. Turner, M. Downey, and M. Dransfield, 2007, First test survey results from the FalconTM helicopter-borne airborne gravity gradiometer system: Preview, **126**, 26–28.

Carcione, J. M., B. Ursin, and J. I. Nordskag, 2007, Cross-property relations between electrical conductivity and the seismic velocity of rocks: Geophysics, **72**, no. 5, E193–E204, http://dx.doi.org/10.1190/1.2762224.

Chen, T., P. Miles, and G. Hodges, 2013, The MULTIPULSE system – high resolution and high power with one TDEM system: 6th International Conference and Exhibition, AEM, Expanded Abstracts.

Chen, T., G., Hodges, A.N., Christensen, and J., Lemieux, 2014, Multipulse airborne TEM technology and test results over oilsands: 76th Annual Conference and Exhibition, EAGE, Workshop, WS14-A02, http://dx.doi.org/10.3997/2214-4609.20140607.

Colombo, D., and M. De Stefano, 2007, Geophysical modeling via simultaneous joint inversion of seismic, gravity and electromagnetic data: application to prestack depth imaging: The Leading Edge, **26**, no. 3, 326–331, http://dx.doi.org/10.1190/1.2715057.

Colombo, D., G.W., McNeice, and R., Ley, 2012, Multigeophysics joint inversion for complex land seismic imaging in Saudi Arabia: 82nd Annual International Meeting, SEG, Expanded Abstracts, http://dx.doi.org/10.1190/segam2012-0441.1.

Colombo, D., G. W. McNeice, D. Rovetta, E. Turkoglu, A. Sena, E. Sandoval-Curiel, F. Miorelli, and Y. Taqi, 2015, Super resolution multi-geophysics imaging of a complex wadi for near-surface corrections: 85th Annual International Meeting, SEG, Expanded Abstracts, 849–853, http://dx.doi.org/10.1190/segam2015-5841055.1.

Gallardo, L. A., and M. A. Meju, 2004, Joint two-dimensional DC resistivity and seismic travel time inversion with cross-gradients constraints: Journal of Geophysical Research, **109**, B3, B03311, http://dx.doi.org/10.1029/2003JB002716.

Habashy, T. M., and A. Abubakar, 2004, A general framework for constraint minimization for the inversion of electromagnetic measurements: Progress in Electromagnetics Research, **46**, 265–312, http://dx.doi.org/10.2528/PIER03100702.

Nabighian, M. N., and J. C. Macnae, 2005, Electrical and EM methods, 1980–2005: The Leading Edge, **24**, no. s1, s42–s45, http://dx.doi.org/10.1190/1.2112391.
Rovetta, D., D. Colombo, and G. W. McNeice, 2013. Spatially constrained transient EM inversion for near-surface characterization in Saudi Arabia, 83rd Annual International Meeting, SEG, Expanded Abstracts, 733–738, http://dx.doi.org/10.1190/segam2013-0484.1.
Sørensen, K. I., and E. Auken, 2004, SkyTEM – a new high-resolution helicopter transient electromagnetic system: Exploration Geophysics, **35**, no. 3, 191–199, http://dx.doi.org/10.1071/EG04194.
Taylor, S., 2005, Direct drilling of VTEM data: Prospectors and developers association of Canada, 73rd annual Convention, Toronto.
Viezzoli, A., A. V. Christiansen, E. Auken, and K. I. Sørensen, 2008, Quasi-3D modeling of airborne TEM data by spatially constrained inversion: Geophysics, **73**, no. 3, F105–F113, http://dx.doi.org/10.1190/1.2895521.
Weijermars, R., 1998, Plio-Quaternary Movement of the East Arabian Block: GeoArabia, **3**, 509–539.